STAT301 Homework 7 Name CHARLES SANDAHL

1. Let A be $A=\begin{pmatrix} 1 & 2 \\ -2 & 1 \end{pmatrix}$ and $B=\begin{matrix} 3 & 5 \\ -1 & 4 \end{matrix}$, find A+B

$$\begin{bmatrix} 1 & 2 \\ -2 & 1 \end{bmatrix} + \begin{bmatrix} 3 & 5 \\ -1 & 4 \end{bmatrix} = \begin{bmatrix} 4 & 7 \\ -3 & 5 \end{bmatrix}$$

2. Let A be $A=\begin{pmatrix} 2 & 0 & -1 \\ 4 & -5 & 2 \end{pmatrix}$ and B be $B=\begin{pmatrix} 1 & 2 \\ 3 & 0 \\ 0 & 1 \end{pmatrix}$, find A*B

$$\begin{bmatrix} 2 & 0 & -1 \\ 4 & -5 & 2 \end{bmatrix} \cdot \begin{bmatrix} 1 & 2 \\ 3 & 0 \\ 0 & 1 \end{bmatrix} = \begin{matrix} 2\cdot 1 + 3\cdot 0 + -1\cdot 0 & 4\cdot 1 + -5\cdot 3 + 2\cdot 0 \\ 2\cdot 2 + 0\cdot 0 + -1\cdot 1 & 4\cdot 2 + -5\cdot 0 + 2\cdot 1 \end{matrix}$$

$$A\cdot B = \begin{bmatrix} 2 & 3 \\ -11 & 10 \end{bmatrix}$$

3. Let A be $A=\begin{matrix} 8 & 6 \\ 5 & 4 \end{matrix}$, find the inverse of A

$$8\cdot 4 - 5\cdot 6$$
$$32 - 30 = 2$$

$$\frac{1}{2}\begin{bmatrix} 4 & -6 \\ -5 & 8 \end{bmatrix} = \begin{bmatrix} 2 & -3 \\ \frac{-5}{2} & 4 \end{bmatrix} = A^{-1}$$

4. Let B be $B=\begin{matrix}1 & 2\\ 4 & 7\end{matrix}$, find the inverse of B if exists

$1\cdot 7 - 4\cdot 2 = -1$

$7 - 8$

$\frac{1}{-1}\begin{bmatrix}7 & -2\\ -4 & 1\end{bmatrix} = \begin{bmatrix}-7 & 2\\ 4 & -1\end{bmatrix} = B^{-1}$

5. Let A be $A=\begin{pmatrix}3 & 0 & 4\\ 2 & 3 & 2\\ 0 & 5 & -1\end{pmatrix}$, find detA

$\begin{bmatrix}3 & 0 & 4\\ 2 & 3 & 2\\ 0 & 5 & -1\end{bmatrix}$

$0 + 1^{\overset{5}{3+2}}\cdot 5\cdot \det\begin{pmatrix}3 & 4\\ 2 & 2\end{pmatrix} + 1^{\overset{6}{3+3}}\cdot -1\cdot \det\begin{pmatrix}3 & 0\\ 2 & 3\end{pmatrix}$

$3\cdot 2 - 2\cdot 4$ $\qquad$ $3\cdot 3 - 0 = 9$

$6 - 8$ $\qquad$ $1^{6}\cdot -1\cdot 9 = -9$

$1^{5}\cdot 5\cdot -2$ $\qquad$ -9

$0 + 10 - 9 = 1$

$\det A = 1$

6. Let A be $A=\begin{pmatrix}3 & 0 & 4\\ 2 & 3 & 2\\ 0 & 5 & -1\end{pmatrix}$, find A^T (A transpose).

$\begin{bmatrix}3 & 2 & 0\\ 0 & 3 & 5\\ 4 & 2 & -1\end{bmatrix} = A^T$

7. Let B be $B=\begin{pmatrix}8 & 1\\ 4 & 2\end{pmatrix}$, find detB

$8\cdot 2 - 4\cdot 1 = 12$ $\qquad$ $\det B = 12$

8. Let B be $B=\begin{pmatrix} 2 & 5 \\ 10 & 7 \end{pmatrix}$, is $v=\begin{pmatrix} 1 \\ 2 \end{pmatrix}$ an eigenvector to B? Why? Show your work.

$$\begin{bmatrix} 2 & 5 \\ 10 & 7 \end{bmatrix}\begin{bmatrix} 1 \\ 2 \end{bmatrix} \qquad \begin{matrix} 2+10 \\ 10+14 \end{matrix} \begin{bmatrix} 12 \\ 24 \end{bmatrix} \quad Bv = 12\begin{bmatrix} 1 \\ 2 \end{bmatrix}$$

yes it is with an eigenvalue of 12

9. Let A be $A=\begin{pmatrix} 5 & 3 \\ 3 & 5 \end{pmatrix}$, find the eigenvalues and eigenvectors of A by hand

$$\begin{bmatrix} 5 & 3 \\ 3 & 5 \end{bmatrix} - \begin{bmatrix} \lambda & 0 \\ 0 & \lambda \end{bmatrix} = \begin{bmatrix} 5-\lambda & 3 \\ 3 & 5-\lambda \end{bmatrix}$$

$$\det\begin{bmatrix} 5-\lambda & 3 \\ 3 & 5-\lambda \end{bmatrix}$$

$$((5-\lambda)(5-\lambda)) - 3\cdot 3$$

$$\lambda^2 - 10\lambda + 25 - 9$$

$$\lambda^2 - 10\lambda + 16$$

$$(\lambda-8)(\lambda-2)$$

$A - \lambda I_x = 0$

$$\begin{bmatrix} 5 & 3 \\ 3 & 5 \end{bmatrix} - \begin{bmatrix} 8 & 0 \\ 0 & 8 \end{bmatrix} = \begin{bmatrix} -3 & 3 \\ 3 & -3 \end{bmatrix}$$

$$\begin{matrix} -3x_1 \; 3x_2 = 0 \\ 3x_1 - 3x_2 = 0 \end{matrix} = \begin{bmatrix} 1 \\ 1 \end{bmatrix}$$

$$\begin{bmatrix} 5 & 3 \\ 3 & 5 \end{bmatrix} - \begin{bmatrix} 2 & 0 \\ 0 & 2 \end{bmatrix} = \begin{bmatrix} 3 & 3 \\ 3 & 3 \end{bmatrix}$$

$$\begin{bmatrix} 3x_1 & 3x_2 \\ 3x_1 & 3x_2 \end{bmatrix} = 0 = \begin{bmatrix} 1 \\ -1 \end{bmatrix}$$

eigen values = 8, 2 eigenvectors = $\begin{bmatrix} 1 \\ 1 \end{bmatrix}, \begin{bmatrix} 1 \\ -1 \end{bmatrix}$

10. Let B be $B=\begin{pmatrix} 7 & -2 \\ 2 & 3 \end{pmatrix}$, find the eigenvalues and eigenvectors of B by hand

$$\begin{bmatrix} 7 & -2 \\ 2 & 3 \end{bmatrix} - \begin{bmatrix} \lambda & 0 \\ 0 & \lambda \end{bmatrix} = \begin{bmatrix} 7-\lambda & -2 \\ 2 & 3-\lambda \end{bmatrix}$$

$$\begin{bmatrix} 7 & -2 \\ 2 & 3 \end{bmatrix} - \begin{bmatrix} 5 & 0 \\ 0 & 5 \end{bmatrix}$$

$$= \begin{bmatrix} 2 & -2 \\ 2 & -2 \end{bmatrix}$$

$$((7-\lambda)(3-\lambda)) - 2 \cdot -2$$

$$-4$$

$$\lambda^2 - 10\lambda + 21 + 4$$

$$\lambda^2 - 10\lambda + 25$$

$$(\lambda - 5)(\lambda - 5)$$

$$2x_1 - 2x_2 = 0$$

$$2x_1 - 2x_2 = 0$$

eigenvectors: $\begin{bmatrix} 1 \\ 1 \end{bmatrix}$

eigenvalues = 5, 5